Supplementary Material: A Label-Replication Artefact, a Measured Leakage Ladder and a Null-Anchored Benchmark for PPG-Based Blood Glucose Estimation

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S1. TRIPOD+AI CHECKLIST

Items of the TRIPOD+AI statement (Collins *et al.*, BMJ 2024) and where each is addressed. Items that do not apply to a benchmark of public data are marked n/a.

Item	Requirement	Where
1	Title identifies study as developing/evaluating a prediction model	Title, Abstract
2	Abstract: objectives, data, methods, results, conclusions	Abstract
3a–b	Background, rationale, objectives	Sec. I
4	Data sources, dates, setting	Sec. III-A, III-C, III-D
5a	Participants: eligibility, recruitment	Sec. III (as reported by dataset authors)
5b–c	Treatment / care pathway	n/a
6	Data preparation: cleaning, windowing, missing data	Sec. IV-A; loaders in code
7	Outcome definition and measurement	Sec. III (fingerstick meter; Dexcom G6)
8	Predictors: definition and measurement	Sec. IV-A, IV-C
9	Sample size justification	Sec. IV-E (minimum detectable effect)
10	Missing-data handling	Sec. IV-A (gap-free windows only)
11	Analytical methods: model type, hyper-parameters, selection	Sec. IV-C, IV-D
12a	Evaluation: splits, protocol, leakage control	Sec. IV-B; frozen folds in code
12b	Model performance measures	Sec. IV-E
12c	Model comparison and null models	Sec. IV-C, IV-E
12d	Uncertainty (CIs, tests)	Sec. IV-E; Tables III–IV; Sec. S4
12e	Fairness / subgroup analysis	Per-subject results (Fig. 5b); sex not analysed (n too small)

13	Class imbalance / outcome distribution	Sec. III; glucose ranges reported
14	Open science: code, data, protocol	Footnote 1; README (incl. encoder weight sources and licences)
15	Patient and public involvement	n/a (public data)
16–17	Participant flow, characteristics	Sec. III; Sec. S3 (cohort table)
18a–f	Model performance, comparisons, uncertainty, subgroups	Sec. V
19	Model updating	n/a
20	Interpretation; comparison with prior work	Sec. VI; Table I
21	Limitations	Sec. VI
22	Usability: inputs, outputs, intended use	Sec. IV-G (intended use cases and their nulls)
23	Funding, conflicts	To be completed by the author

S2. Model information sheet

Field	**Value**
Task	Regression of blood / interstitial glucose (mg/dL) from a 5-s PPG window (networks, classical) or 10–30-s segments (frozen encoders)
Intended use	Benchmark only; no clinical use
Inputs	320 samples at 64 Hz, 0.5–8 Hz bandpassed, z-scored per window (variants in Sec. V-G)
Outputs	One glucose value per window; averaged per reference measurement
Training data	Kermani–Esmaeili recordings (LOSO folds); BIG IDEAs training periods; STUBA ePPG (LOSO)
Hyper-parameters	AdamW, lr 10^{-3}, wd 10^{-4}, one-cycle, 40/15 epochs, batch 16/64, grad-clip 1, smooth-L1 on standardised target
Seeds	Kermani–Esmaeili networks: 3; STUBA 1-D CNN: 3, other STUBA networks one; BIG IDEAs residual 1-D CNN: 3, other BIG IDEAs rows one; encoder probes and classical models are deterministic (one run)
Null models	Train mean; subject train mean; time of day; persistence
Leakage controls	Named protocols with asserted guarantees; frozen folds; guard gap; full-file label-construction test
Known failure modes	Predicts near the (subject) mean; no event sensitivity

Pretrained encoders	PaPaGei-S (5.8 M, 125 Hz, 10-s), Pulse-PPG (28.5 M, 50 Hz, 10–30-s), AnyPPG (4.0 M, 125 Hz, 10-s); frozen released weights loaded weights-only; 0.5–12 Hz Chebyshev-II, resample, z-score; 512-d embeddings averaged per reference; RidgeCV ($\alpha \in [10^{-2}, 10^{4}]$) on training-fold-standardised features
Licences	Data: Kermani–Esmaeili CC BY 4.0 (Mendeley Data); BIG IDEAs PhysioNet ODC-By; STUBA ePPG CC0. Encoders: PaPaGei code BSD-3-Clause, weights Zenodo 13983110 (no licence stated); Pulse-PPG MIT (code and weights, Zenodo 17345536); AnyPPG checkpoint from GitHub PKUDigitalHealth/AnyPPG (no licence file)

S3. COHORT CHARACTERISTICS

	Kermani–Esmaeili	**BIG IDEAs**	**STUBA ePPG**
Subjects	23	16	24
Male / female	15 / 8	7 / 9	20 / 4
Age (years)	22–61	35–65 (inclusion criterion)	22–48
Diabetes	not stated	HbA1c 5.3–6.4%	3 T1DM
Reference	Accu-Chek Active	Dexcom G6 CGM	Beurer GL49
Reference measurements	67	28867	125
Glucose range (mg/dL)	88–183	40–261	40–304
< 70 / > 180 mg/dL	0 / 2	186 / 431	34 / 18
Sensor	finger, 550 nm, 2175 Hz	wrist E4 BVP, 64 Hz	finger, 530/655/940 nm, 100 Hz

S4. ALL PAIRED COMPARISONS AGAINST THE NULL

Every model-versus-null comparison in the paper (41 in total; $\star$: the pre-specified primary comparison, all others exploratory). MAE of the (seed-ensemble) prediction in mg/dL; Δ: paired MAE difference against the cohort's null (train mean; subject training-period mean on BIG IDEAs) with its 95% subject-cluster bootstrap interval. "better"/"worse": interval entirely below/above zero. 2 intervals lie below zero and 16 above; no correction for multiplicity is applied.

Cohort	**Model**	**MAE**	**Δ**	**95% CI**	**Verdict**
Kermani–Esmaeili	Ridge, PPG	14.62	-0.09	[-0.64, +0.62]	–
Kermani–Esmaeili	Ridge, demographics	15.57	+0.86	[-0.32, +2.27]	–
Kermani–Esmaeili	Ridge, PPG + demogr.	15.23	+0.52	[-1.17, +2.57]	–
Kermani–Esmaeili	GBM, PPG	16.61	+1.89	[+0.15, +3.47]	worse
Kermani–Esmaeili	GBM, PPG + demogr.	15.49	+0.78	[-1.28, +2.99]	–
Kermani–Esmaeili	1-D CNN$\star$	13.31	-1.41	[-2.99, -0.03]	better
Kermani–Esmaeili	2-D CNN (CWT)	13.91	-0.81	[-2.51, +0.75]	–
Kermani–Esmaeili	PointNet (Takens)	14.01	-0.70	[-1.83, +0.38]	–
Kermani–Esmaeili	1D+2D+3D fusion	14.66	-0.05	[-1.48, +1.25]	–
Kermani–Esmaeili	ResNet-1D	15.01	+0.30	[-1.42, +2.01]	–
Kermani–Esmaeili	CNN–LSTM	13.34	-1.37	[-2.95, +0.23]	–
Kermani–Esmaeili	PaPaGei-S probe	14.75	+0.03	[-0.91, +1.05]	–
Kermani–Esmaeili	AnyPPG probe	14.91	+0.19	[-0.17, +0.67]	–
Kermani–Esmaeili	Pulse-PPG probe	14.21	-0.50	[-1.67, +0.79]	–
BIG IDEAs	Ridge, PPG	17.32	+2.04	[+0.66, +3.83]	worse
BIG IDEAs	1-D CNN	16.94	+1.66	[+0.48, +3.11]	worse
BIG IDEAs	2-D CNN (CWT)	16.97	+1.69	[+0.50, +3.18]	worse

BIG IDEAs	PointNet (Takens)	16.98	+1.70	[+0.48, +3.22]	worse
BIG IDEAs	1D+2D+3D fusion	16.68	+1.40	[+0.22, +2.85]	worse
BIG IDEAs	Time of day + subject	15.29	+0.01	[-0.49, +0.53]	–
BIG IDEAs	Persistence	31.35	+16.07	[+3.83, +29.92]	worse
BIG IDEAs	Ridge, PPG, residual	15.13	-0.15	[-0.36, +0.09]	–
BIG IDEAs	1-D CNN, residual	15.04	-0.24	[-0.70, +0.23]	–
BIG IDEAs	2-D CNN (CWT), residual	15.09	-0.19	[-0.71, +0.31]	–
BIG IDEAs	PointNet (Takens), residual	15.04	-0.24	[-0.69, +0.17]	–
BIG IDEAs	1D+2D+3D fusion, residual	15.07	-0.21	[-0.73, +0.31]	–
BIG IDEAs	PaPaGei-S probe	17.24	+1.96	[+0.61, +3.68]	worse
BIG IDEAs	AnyPPG probe	16.79	+1.51	[+0.29, +3.20]	worse
BIG IDEAs	Pulse-PPG probe	16.99	+1.70	[+0.38, +3.41]	worse
BIG IDEAs	PaPaGei-S probe, residual	15.07	-0.21	[-0.46, +0.04]	–
BIG IDEAs	AnyPPG probe, residual	15.08	-0.21	[-0.58, +0.16]	–
BIG IDEAs	Pulse-PPG probe, residual	15.01	-0.27	[-0.54, -0.02]	better
STUBA ePPG	Ridge, PPG	44.39	+0.89	[+0.35, +1.22]	worse
STUBA ePPG	GBM, PPG	48.33	+4.83	[+1.76, +9.16]	worse
STUBA ePPG	1-D CNN	49.55	+6.05	[+2.35, +7.65]	worse
STUBA ePPG	2-D CNN (CWT)	48.61	+5.10	[+2.70, +6.32]	worse
STUBA ePPG	PointNet (Takens)	43.20	-0.30	[-1.28, +1.95]	–
STUBA ePPG	1D+2D+3D fusion	47.52	+4.01	[+1.30, +5.16]	worse
STUBA ePPG	PaPaGei-S probe	43.93	+0.43	[-0.01, +1.75]	–
STUBA ePPG	AnyPPG probe	43.51	+0.01	[-0.75, +1.08]	–
STUBA ePPG	Pulse-PPG probe	44.58	+1.08	[+0.07, +1.88]	worse